

Red Zone Risk: Forecasting NFL Player Injury Likelihood



Alec Morris

April '26

Problem Statement

- Why do we care about injury likelihood?
 - Career longevity and post-playing outlook
 - Team performance/financial benefits
 - Preventative measures that mitigate these risks may exist

Research Question

- Can we use existing data to help predict and prevent player injuries?
 - What data points will be pertinent?
 - How much data is available publicly that will aid us?

Problem Statement

- Why do we care about injury likelihood?
 - Career longevity and post-playing outlook
 - Team performance/financial benefits
 - Preventative measures that mitigate these risks

Data Sources, Methods, and Plan



Data Sources

- Nflreadpy will be primary source of relevant info
- This may need to be supplemented by other sources as we move forward- NFL.com , etc.

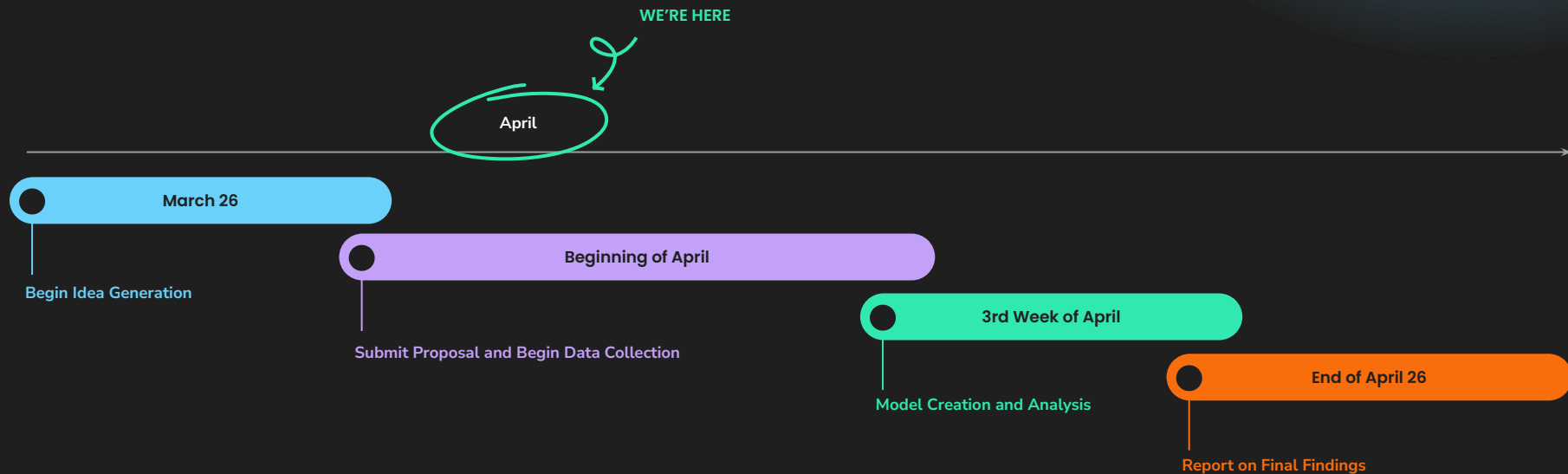
Methodology

We may need to look in a few directions here, but generally I plan to use older available data to train a model and predict outcomes on newer available data.

Plan

End up with a model that is testable and both precise and accurate. Once there's a better idea of the data, we can lock down exactly what the acceptable parameters would be.

Timeline



Outcomes

- What do we expect to learn from this experience?
 - Associable scores that demonstrate injury likelihood
 - Correlation analysis between player age, position, and other variables.